

Distilling TabPFN into Lightweight Tabular Models: A Comparison of Distillation Strategies and Where They Win

We distil the in-context tabular foundation model TabPFN into small, cheap-to-serve students (fine-tuned MLPs, augmented Random Forests) and ask, per method, *where* the student is worth deploying. Each track is compared against its own baseline on the project’s suite of small, non-linear OpenML datasets (`StratifiedShuffleSplit`, `random_state=0`). Scores are **normalized ROC-AUC**, $\text{nAUC} = 2 \text{AUC} - 1$ (chance = 0, perfect = 1). The per-track tables below list the five sample datasets and how the model improves over it’s baseline, $\Delta = \text{nAUC}_{\text{distilled}} - \text{nAUC}_{\text{baseline}}$, plus its single worst dataset; the TabPFN teacher is shown for reference.

Embedding student (config 0.2 of dataset)

The strongest distilled model: soft targets beat the baseline by the widest and most consistent margins, and on `monks-problems-2` the student nearly matches the teacher.

Table 1: Soft-label student (config 0.2): datasets gain over the baseline, plus the single worst dataset (normalized ROC-AUC).

Dataset	Classes	Teacher	Distilled	Baseline	Δ
titanic	2	0.691	0.685	0.541	+0.144
haberman	2	0.336	0.273	0.133	+0.140
datatrieve	2	0.510	0.425	0.299	+0.126
monks-problems-2	2	1.000	0.979	0.871	+0.107
megawatt1	2	0.466	0.353	0.252	+0.101
<i>Worst case</i>					
ilpd	2	0.466	0.202	0.412	-0.210

Embedding student (config 8 training examples)

At parity with the baseline overall, but on these datasets soft targets transfer the teacher’s ranking clearly – on `fertility` the student even beats the teacher.

Table 2: Embedding student (config 8): datasets gain over baseline, plus the single worst dataset (normalized ROC-AUC).

Dataset	Classes	Teacher	Distilled	Baseline	Δ
fertility	2	0.058	0.233	0.135	+0.097
datatrieve	2	0.339	0.303	0.213	+0.090
blood-transfusion	2	0.189	0.262	0.185	+0.077
ionosphere	2	0.583	0.476	0.407	+0.069
biomed	2	0.810	0.733	0.706	+0.027
<i>Worst case</i>					
backache	2	0.353	0.022	0.180	-0.159

Distilled MLP (TabDistil method) (config 64 training examples)

The hardest regime, yet distillation still beats an identically sized plain MLP – and on several sets the baseline scores near chance while the distilled model stays clearly positive.

Random Forest with augmentation distillation (config 64 training examples)

The best tree-based student; this table isolates the gain from *augmentation* alone – the datasets where augmented synthetic samples most help a forest over the raw synthetic set.

Table 3: Distilled MLP (config 64): datasets gain over the baseline, plus the single worst dataset (normalized ROC-AUC).

Dataset	Classes	Teacher	Distilled	Baseline	Δ
ar4	2	0.248	0.100	-0.091	+0.191
monks-problems-1	2	0.172	0.139	0.060	+0.078
planning-relax	2	-0.012	0.062	-0.013	+0.075
diabetes	2	0.370	0.008	-0.064	+0.072
tae	3	0.129	0.082	0.049	+0.033
<i>Worst case</i>					
vote	2	0.923	0.254	0.919	-0.665

Table 4: Augmented-RF distillation (config 64): datasets gain over the baseline, plus the single worst dataset (normalized ROC-AUC).

Dataset	Classes	Teacher	Distilled	Baseline	Δ
titanic	2	0.807	0.776	0.768	+0.008
monks-problems-2	2	0.926	0.276	0.222	+0.054
tic-tac-toe	2	0.862	0.635	0.594	+0.041
climate-crashes	2	0.362	0.252	0.220	+0.031
titanic	2	0.615	0.553	0.538	+0.015
<i>Worst case</i>					
ar1	2	0.393	0.244	0.404	-0.159

MLP distillation methods compared

We evaluate two ways to distil TabPFN into an MLP and seven variants in total. **Embedding student** has two main parts: encoder and decoder. First we train the encoder (simple MLP) to predict a train example’s embeddings based on the TabPFN’s embeddings. At the same time we train a decoder head (Linear Layer) to predict a class from embeddings. **Plain MLP** (baseline) trains directly on the real labels (no teacher; reference). **Distilled MLP** uses aggregated embeddings of the training examples, then uses linear transform to create MLP weights. **Distilled + finetuned** additionally runs a few epochs on the ground-truth hard labels and synthetic data from the TabPFN. **Random Forest with augmentation** is a simple Random Forest fitted on synthetic data generated by randomly augmenting training data and making predictions with TabPFN.

Takeaways. Distillation pays off selectively, not uniformly. The augmented-RF and config-02 students recover most of TabPFN’s ranking quality and win broadly over their baselines, making them the recommended deployable surrogates. For MLP students, the gains come from *augmented synthetic data* rather than from soft targets on the original inputs; plain real-data KD does not, on average, beat a vanilla MLP. Even the weaker students still post clear, dataset-specific wins (Tables 1–4), so the practical recommendation is to pick the student per dataset using these margins.

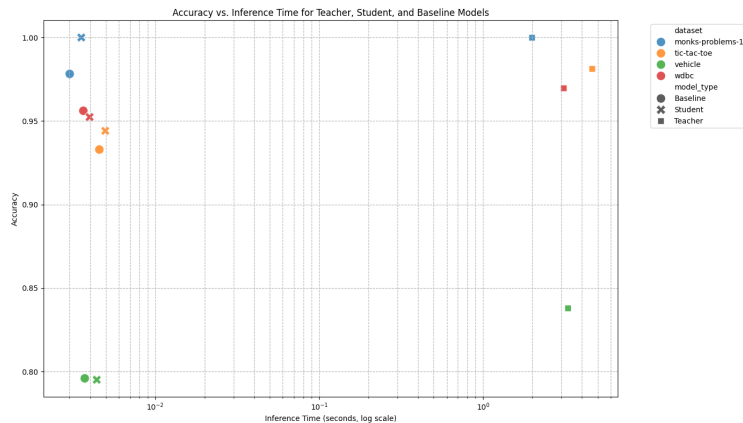


Figure 1: Inference time vs accuracy